Vision–Based Water Surface Velocimetry from Moving Cameras

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Abstract

Estimating surface water velocity from video is a key requirement in hydrology and flood monitoring. Classical image-based velocimetry methods such as LSPIV and optical flow assume a static camera, an assumption that can not hold for mobile platforms where camera rotation and drift distort the observed motion field. This work investigates further into whether meaningful water-surface velocity estimates can be recovered from monocular video captured by a moving camera.

This study present a modular processing pipeline that combines planar homography rectification, region-of-interest masking, and robust sparse feature tracking with IMU-assisted orientation stabilization. Raw gyroscope, accelerometer, and magnetometer measurements are fused using an AHRS filter to estimate camera orientation, and remove camera motion prior to optical flow estimation. Velocity vectors are converted from pixel displacements to metric units using scale calibration.

Compensating for camera rotation significantly improves the stability of optical velocity estimates under motion. The resulting system provides a baseline for vision-based water-surface velocimetry from moving platforms and establishes a foundation for future extensions incorporating translational motion.

1 Introduction

Surface water velocity is a key variable in hydrology and water resources engineering, supporting discharge estimation, flood monitoring, and system-level management of rivers and irrigation networks Dingman [2015], Rantz [1982]. Conventional discharge and velocity measurement methods are accurate but often require direct access, specialized instrumentation (such as ADCP's), or fixed installations, which can be difficult to deploy at scale or during hazardous flood conditions Rantz [1982].

Image-based velocimetry offers a alternative by estimating surface motion from video of tracers such as foam, debris, or naturally occurring texture, without the need of planted tracers. In hydraulic engineering, Large-Scale Particle Image Velocimetry (LSPIV) and related PIV/PTV techniques have been demonstrated for flow analysis Fujita et al. [1998], Creutin et al. [2003], Muste et al. [2008]. Practical fixed-camera workflows typically rely on geometric rectification and pixel-to-metric scaling to map image-plane displacements into physical velocity units Patalano et al. [2017].

A core limitation of many existing deployments is the assumption of a static camera pose. When the camera moves, our image beecomes noisy and unreliable due to drift, violating the conditions for classical methods such as LSPIV/PTV. This issue is especially relevant for mobile and rapid-deployment settings, where imagery may be collected from handheld devices, UAVs, or floating platforms Dramais et al. [2011], Eltner et al. [2020], Perks [2020].

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In this work, we study a reproducible monocular pipeline for water-surface velocimetry under moving-camera conditions. The pipeline combines planar rectification with robust sparse feature tracking based on the Lucas–Kanade framework Lucas and Kanade [1981] and adopts processing choices consistent with existing image velocimetry toolchains Patalano et al. [2017], Perks [2020]. Our goal is not to solve camera motion estimation, but to highlights failure modes, practical calibration steps, and measurable improvements when motion compensation is incorporated.

2 Literature Review

2.1 Conventional Velocity Measurement

Classical approaches to streamflow measurement rely on in-situ instruments such as current meters, acoustic Doppler current profilers (ADCPs), and tracer-based techniques Rantz [1982], Dingman [2015]. These methods provide accurate point or cross-sectional measurements. However, they require direct access to the channel and specialized equipment which severly limits their deployment in hazardous flood conditions or remote environments.

These practical constraints have motivated the exploration of non-contact measurement techniques.

2.2 Image-Based Velocimetry in Riverine Environments

Image-based velocimetry is a promising alternative for measuring surface flow velocity by analyzing visible tracers on the water surface. Large-Scale Particle Image Velocimetry (LSPIV) adapts classical PIV methods to outdoor hydraulic environments, enabling surface velocity estimation using standard video cameras Fujita et al. [1998], Muste et al. [2008]. Early studies demonstrated that LSPIV could be used to produce velocity estimates consistent with traditional measurements Creutin et al. [2003].

In these approaches, surface tracers are identified and tracked across consecutive frames, and pixel displacements are converted to physical velocities through geometric calibration. The effectiveness of image-based velocimetry depends on sufficient surface texture, stable illumination, and well-defined camera geometry. When these conditions are satisfied, LSPIV and related methods are much more effective.

2.3 Geometric Rectification and Fixed-Camera Toolchains

To map image-plane motion into real-world coordinates, most image-based velocimetry pipelines employ geometric rectification using planar homographies Hartley and Zisserman [2003]. This process assumes that the observed water surface can be approximated as a plane and that the camera pose remains fixed. Under these assumptions, a single homography can be used to convert pixel displacements into metric units.

Several softwares have been developed to standardize this workflow for practical use. The RIVeR toolbox provides an integrated framework for homography estimation, scale calibration, and velocity post-processing, facilitating large-scale deployment of image-based velocimetry under fixed-camera conditions Patalano et al. [2017]. Such systems have contributed significantly to the adoption of image velocimetry in rivers and irrigation channels. However, these techniques can not keep up in scenarios in which the camera undergoes motion.

2.4 Mobile Platforms and Moving-Camera Image Velocimetry

Recent work has explored the use of mobile platforms, and unmanned aerial vehicles, for river monitoring and flood discharge measurement Dramais et al. [2011], Eltner et al. [2020]. Mobile platforms offer improved spatial coverage and rapid deployment but introduce significant challenges for image-based velocimetry. Camera motion produces apparent image displacements that are often larger than the true surface motion, leading to biased velocity estimates if not properly compensated.

Perks [2020] introduced a flexible image velocimetry framework capable of operating on both fixed and mobile platforms, highlighting the importance of robust feature tracking and careful motion handling. While such approaches extend classical methods to mobile settings, they also emphasize that moving-camera scenarios remain significantly more challenging than fixed-camera deployments, particularly in environments with weak texture and non-rigid surface appearance, issues that make feature tracking almost impossible.

2.5 Feature Tracking and Optical Flow Methods

Sparse feature tracking methods based on the Lucas–Kanade formulation remain widely used due to their computational efficiency and robustness to moderate noise Lucas and Kanade [1981]. By tracking distinct image features across frames, these methods provide local displacement vectors that can be aggregated into velocity estimates.

The performance of feature tracking is strongly influenced by camera motion. Global rotations introduce coherent optical flow patterns that can overwhelm the local motion induced by water surface flow. As a result, even robust tracking algorithms produce misleading velocity estimates when applied directly to moving-camera footage.

2.6 Summary and Identified Research Gap

The reviewed literature demonstrates that image-based velocimetry is a well-established technique for surface flow measurement under fixed-camera conditions. Geometric rectification and feature-based tracking pipelines have been successfully applied in rivers and irrigation channels. However, the assumption of a static camera is a constant in most existing approaches.

Although mobile platforms have been explored, systematic treatment of camera motion—particularly rotational motion—remains limited in practical velocimetry pipelines. This gap motivates the present work, which focuses on a complete and reproducible implementation of feature-based surface velocity estimation augmented with explicit handling of camera rotation, providing a practical baseline for moving-camera image velocimetry.

3 Problem Formulation

Image-based surface velocimetry estimates water velocity by measuring apparent motion in the image plane between consecutive video frames. When the camera is fixed relative to the scene, pixel displacements are caused solely by the motion of surface tracers and can be mapped to physical velocity. However, this assumption breaks down completely for cameras mounted on floating or drifting platforms.

3.1 Observed Image Motion

Let $\mathbf{x}_t \in \mathbb{R}^2$ denote the image-plane position of a surface feature at time t. The observed inter-frame displacement $\Delta\mathbf{x}_{\text{obs}}$ between frames t and $t + \Delta t$ can be expressed as

$$\Delta\mathbf{x}_{\text{obs}} = \Delta\mathbf{x}_{\text{water}} + \Delta\mathbf{x}_{\text{camera}}, \tag{1}$$

where $\Delta\mathbf{x}_{\text{water}}$ represents the true displacement of the water surface feature, and $\Delta\mathbf{x}_{\text{camera}}$ represents the apparent displacement induced by camera motion.

For floating camera nodes, $\Delta\mathbf{x}_{\text{camera}}$ arises from a combination of platform translation, rotation (yaw, pitch, roll), and high-frequency jitter due to waves. In practice, even small angular rotations produce large pixel displacements, often exceeding those caused by the water flow itself.

3.2 Limitations of Classical Optical Velocimetry

Classical optical velocimetry methods such as LSPIV, sparse feature tracking, and dense optical flow implicitly assume

$$\Delta\mathbf{x}_{\text{camera}} \approx \mathbf{0}. \tag{2}$$

Under this assumption, the observed displacement can be directly interpreted as water motion. When the camera moves, this assumption is violated, causing these methods to fail catastrophically. The estimated velocity field becomes biased and spatially inconsistent, rendering physical interpretation invalid.

3.3 Geometric Rectification and Planar Assumptions

Most practical image velocimetry pipelines operate under the assumption that the water surface is approximately planar. This allows pixel coordinates to be mapped into a rectified, top-down reference frame using a planar projective transformation (homography). After rectification, pixel displacements become approximately uniform across the image, enabling conversion from pixels to meters using a single scale factor.

However, geometric rectification alone does not address camera motion over time. A single homography computed from a reference frame is only valid if the camera pose remains fixed. For a moving camera, the effective homography becomes time-varying, and failure to account for this variation leads to systematic errors in velocity estimation.

3.4 Problem Statement

The central problem addressed in this work can therefore be stated as follows:

> *Given monocular video captured by a camera undergoing unknown motion, how can true water-surface velocity be recovered in physical units using image-based techniques?*

3.5 Scope and Assumptions

This work focuses on correcting camera *orientation* (rotation) rather than full six-degree-of-freedom camera motion. Translational motion parallel to the water surface is assumed to be either small relative to flow velocity. This work establish a reproducible and interpretable baseline for moving-camera surface velocimetry. The odometry of the moving platform itself and the effects of translational motion can be corrected by implementing gnss tracking or onboard odometry techniques to find the displacement of the camera.

These assumptions allow the problem to remain tractable.

4 Simulation Setup and Synthetic Data Generation

4.1 Motivation for a Synthetic Benchmark

Evaluating water-surface velocimetry methods using real-world video is challenging due to the lack of dense, time-synchronized ground truth velocity measurements. In field deployments, reference instruments such as Acoustic Doppler Current Profilers (ADCPs) or current meters typically provide sparse or depth-averaged measurements, making direct validation of image-based surface velocity estimates difficult.

To enable quantitative and repeatable evaluation, we adopt a synthetic benchmarking approach using a numerically generated river velocity field paired with a rendered video sequence. This allows direct comparison between estimated surface velocity and known ground truth at every time step.

4.2 Synthetic Dataset Description

The synthetic dataset consists of a time-indexed sequence of two-dimensional velocity fields defined over a regular spatial grid. We use the synthetic river flow videos and accompanying reference velocity fields provided by Bodart et al. [2022]. For each timestep k, a reference file `ref_velXXXX.dat` provides:

- $X(x_i)$: horizontal spatial coordinates (m),
- $Y(y_j)$: vertical spatial coordinates (m),
- $V_x(x_i, y_j)$: horizontal velocity component (m/s),
- $V_y(x_i, y_j)$: vertical velocity component (m/s).

These velocity fields represent physically plausible open-channel flow patterns and serve as ground-truth surface velocity. A corresponding RGB video sequence depicts advected surface texture consistent with the underlying velocity field.

4.3 Floating-Camera Simulation

To emulate a realistic sensing platform such as a floating node or vessel-mounted camera, we simulate a virtual moving camera that observes only a local region of the river surface.

4.3.1 Camera Translation Model

Let $\mathbf{v}_w = (v_x, v_y)$ denote the local water velocity sampled from the ground-truth field. The camera translation velocity is defined as:

$$\mathbf{v}_{cam} = \alpha \, \mathbf{v}_w, \tag{3}$$

where $\alpha \in [0, 1]$ is a configurable follow gain. For the validation experiments reported in this work, we set $\alpha = 0$, corresponding to a stationary camera. This isolates estimator performance from platform motion.

Camera position is updated via forward Euler integration:

$$\mathbf{p}_{cam}(k+1) = \mathbf{p}_{cam}(k) + \mathbf{v}_{cam}(k) \, \Delta t, \tag{4}$$

where Δt is the video frame period.

4.3.2 Camera Rotation Model

To simulate platform yaw disturbances caused by flow alignment, the camera is rotated according to the local flow direction:

$$\theta_{yaw} = \text{clip} \left(\gamma \cdot \arctan 2(v_y, v_x), \, \pm \theta_{max} \right), \tag{5}$$

where γ is a yaw gain parameter and θ_{max} limits excessive rotation. This rotation is applied via an affine image warp prior to cropping.

4.3.3 Cropping and Output Generation

At each timestep, a fixed-size window (960×540 pixels) centered on the virtual camera position is extracted from the warped frame and written to a new video. This cropped video constitutes the input to the velocity estimation pipeline.

Simultaneously, the true water velocity at the camera location is logged to a CSV file, providing time-aligned ground truth for evaluation.

4.4 Optical Flow–Based Velocity Estimation

Surface velocity is estimated from the floating-camera video using a Lucas–Kanade (LK) feature tracking approach.

At each frame:

1. Shi–Tomasi corner features are detected within the image interior.
2. Features are tracked to the next frame using pyramidal LK optical flow.
3. Per-feature displacement vectors are computed.
4. The top fraction of fastest, directionally consistent tracks is retained.
5. A robust median flow vector is computed and converted to physical units.

Pixel displacement is converted to meters per second via:

$$v_{est} = \|\Delta \mathbf{p}\| \cdot f_{fps} \cdot s_{m/px}, \tag{6}$$

where f_{fps} is the frame rate and $s_{m/px}$ is the calibrated meters-per-pixel scale accounting for cropping and domain geometry.

4.5 Temporal Smoothing and Outlier Rejection

Raw per-frame optical flow estimates are inherently noisy due to feature dropout, illumination variation, and local texture ambiguity. To obtain a physically meaningful velocity signal, two stabilization steps are applied.

Outlier rejection: Estimates exhibiting implausible frame-to-frame jumps are discarded.

Temporal filtering: A first-order exponential moving average is applied:

$$\hat{v}_k = (1 - \beta)\, \hat{v}_{k-1} + \beta\, v_{est,k}, \tag{7}$$

with smoothing coefficient $\beta = 0.2$. This filtering approximates a low-pass.

4.6 Evaluation Metrics

Estimated surface velocity is compared against the ground-truth velocity magnitude at each timestep. Performance is quantified using mean absolute error (MAE) and root mean square error (RMSE).

For the reported experiment:

$$\text{MAE} = 0.280 \text{ m/s}, \quad \text{RMSE} = 0.361 \text{ m/s}. \tag{8}$$

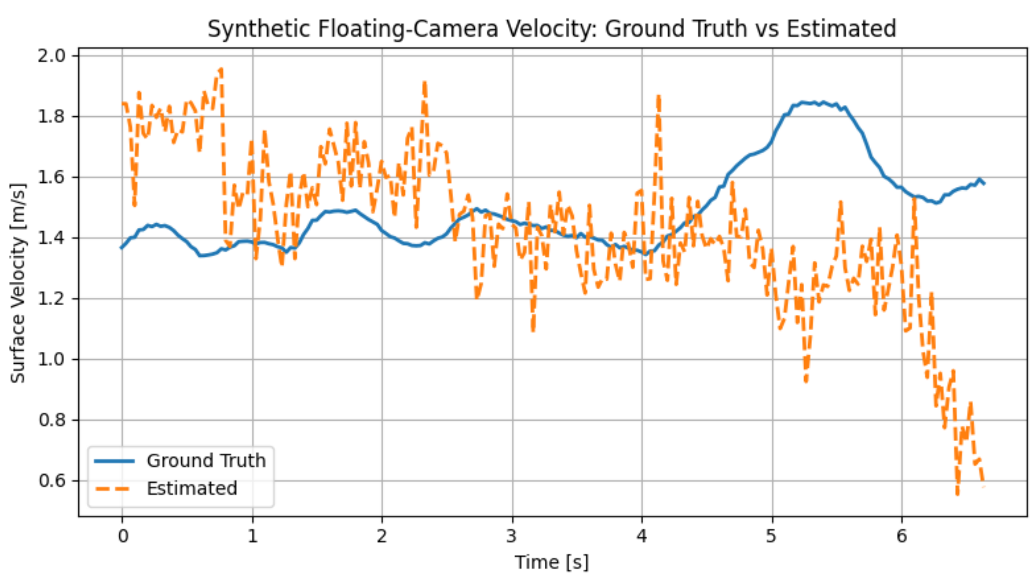

Figure 1: Example frame from the synthetic river flow dataset of Bodart et al. [2022] and the corresponding ground-truth surface velocity magnitude. The last second of data should be discarded.

4.7 Discussion of Simulation

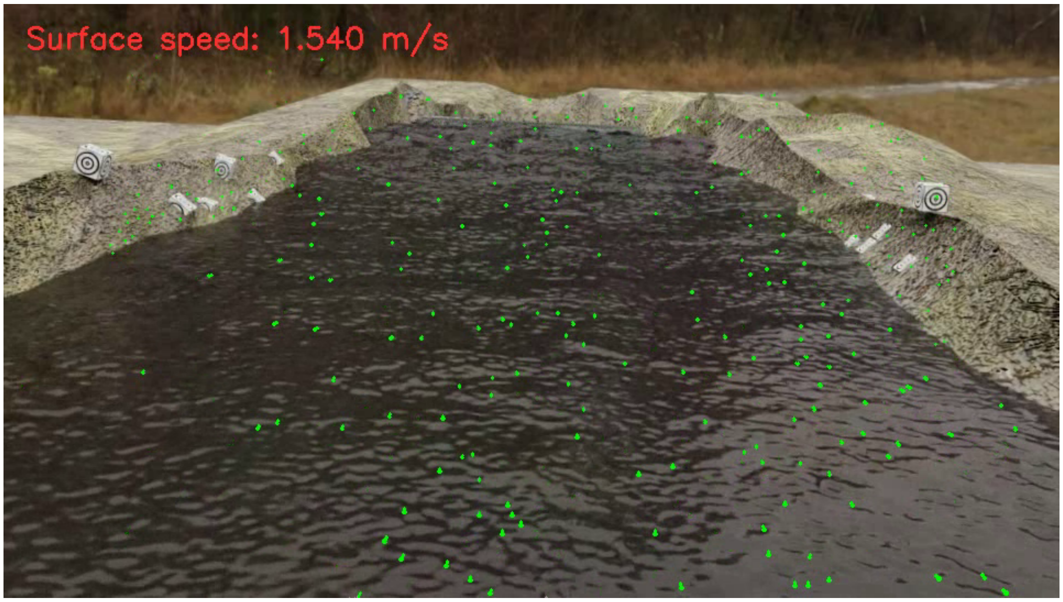

Figure 2: Floating-camera simulation. A virtual camera applies flow-aligned yaw rotation and extracts a fixed-size crop centered on the camera position.

While the synthetic setup provides dense and repeatable ground truth, several limitations remain. Optical flow estimates are sensitive to surface texture realism, and the ground-truth velocity field may not perfectly correspond to the visually advected tracers present in the rendered video. Additionally, the simulated camera does not model real imaging artifacts such as motion blur or rolling shutter effects, which can further degrade optical flow accuracy in real deployments.

The simulation environment also highlights a fundamental challenge inherent to image-based velocimetry: the conversion from pixel displacement to physical velocity. Surface speed is estimated by converting optical flow measurements from pixels per frame to meters per second via a meters-per-pixel scale factor. This scale depends on the assumed correspondence between image pixels and physical distances in the scene, which in practice is influenced by camera altitude, viewing geometry, cropping, and perspective effects.

To address this issue, a scalar calibration factor is applied to align the image-based velocity estimates with the reference velocity field. Our simulation calibration could only be done after viewing ground truth. In real-world deployments, an analogous calibration step would be performed, with additional sensors to help define the ground truth.

5 Real world pipeline and testing

5.1 Calibration and Preprocessing

To convert image-space motion into physically meaningful surface velocity estimates, a multi-stage calibration procedure was performed. This calibration was necessary to (i) restrict analysis to the water surface, (ii) mitigate perspective distortion caused by the camera viewpoint, and (iii) convert pixel displacements into real-world units.

5.2 Region of Interest Selection

A polygonal region of interest (ROI) was manually selected on the input video frame to isolate the visible water surface. This step ensures that optical flow is computed only on waterborne tracers such as foam, ripples, or bubbles, while excluding static background elements such as river banks or surrounding terrain.

Restricting feature tracking to the ROI reduces the influence of static scene elements and improves the robustness of the estimated flow field. An example of the selected ROI is shown in Figure 3.

For the purpose of this video the bottom left of the screen was where most of the features we wished to track lied.

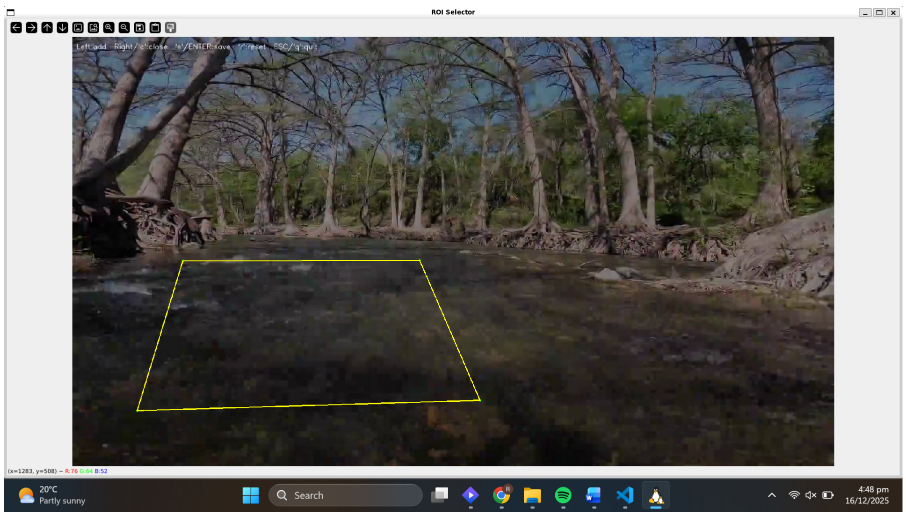

Figure 3: Manual selection of the region of interest (ROI) corresponding to the water surface.

5.3 Planar Homography Rectification

Due to perspective effects introduced by the camera viewpoint, pixel displacements in the raw image do not correspond uniformly to physical distances across the scene. To reduce this effect, a planar homography was estimated using four manually selected points assumed to lie on the water surface plane. This was done largely through trial and error until our warped roi did not look correct.

The homography maps the selected quadrilateral region in the original image to a rectified view, approximating a top-down representation of the river surface. This rectification enables subsequent motion analysis to be performed in a geometrically consistent coordinate system. Figure 4 illustrates the homography point selection and the resulting warped image.

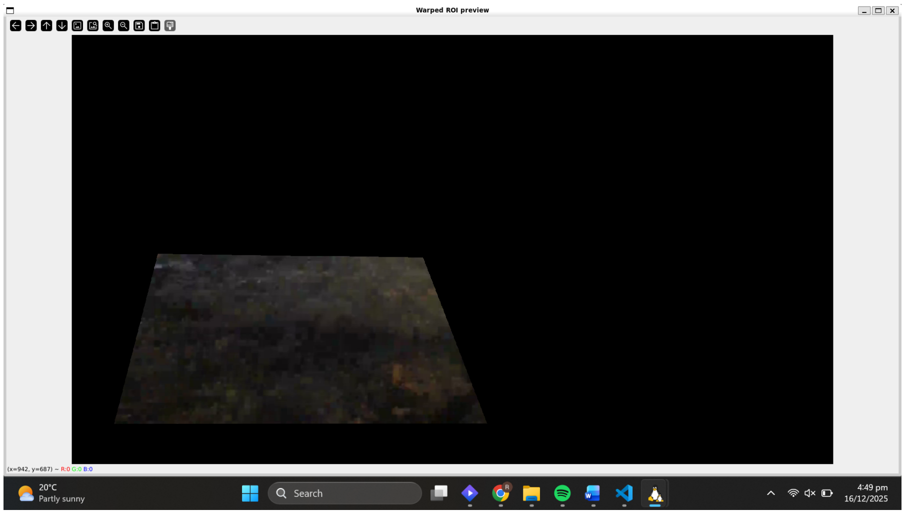

Figure 4: Homography calibration. Four planar points are selected in the original image and mapped to a rectified view. After which our selected roi is warped to match

5.4 Scale Calibration

Following rectification, a pixel-to-meter scale factor was computed by selecting two points in the warped image with a known real-world separation. This scale factor enables conversion of pixel velocities into physical units of meters per second. This can be done by estimating the distance between 2 known points such as the river banks, as was done in this experiment.

This approach provides only an approximate scale, and for the purposes of this experiment the distance between the banks was chosen as 25 meters, though in practical applications knowledge of the deployment location would make this approximation more precise.

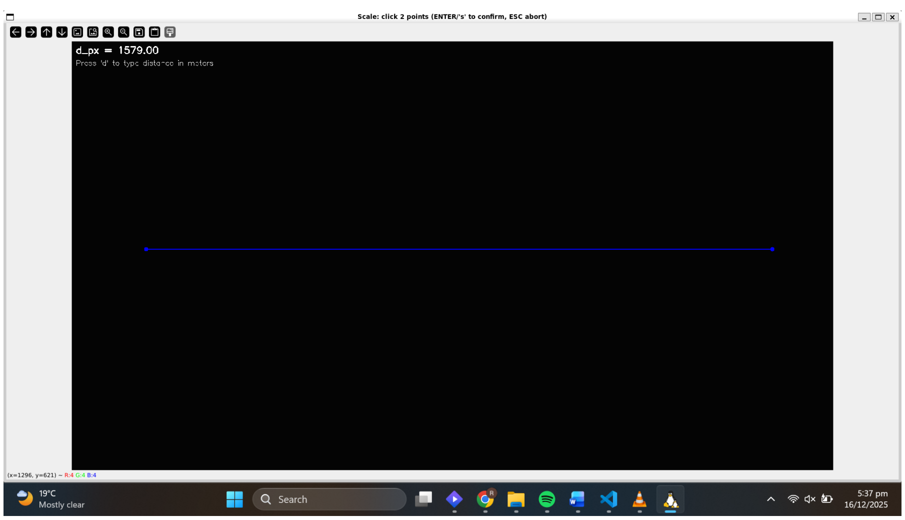

Figure 5: Scale calibration using a known-distance baseline selected in the rectified image. The image taken during the actual calibration process was lost but this image shows it in action

5.5 Key Mathematical Components

This pipeline relies on three core mathematical ideas to convert video motion into surface velocity estimates: planar projective geometry (homography), optical-flow-based feature tracking, and robust model fitting via RANSAC.

Planar Homography Rectification. Assuming the observed river surface is approximately planar, pixel coordinates in the image are mapped into a rectified (top-down) plane using a 3×3 homography matrix H:

$$\tilde{\mathbf{x}}' \sim H\tilde{\mathbf{x}}, \tag{9}$$

where $\tilde{\mathbf{x}} = [x\ y\ 1]^T$ are homogeneous pixel coordinates and $\sim$ denotes equality up to scale. After rectification, spatial scaling becomes approximately uniform within the region of interest (ROI), enabling a single pixel-to-meter scale factor to be applied.

Optical Flow and Velocity Conversion. Surface tracers are tracked frame-to-frame using sparse Lucas–Kanade tracking. For each tracked feature, the pixel displacement is

$$\Delta\mathbf{p}_{px} = \mathbf{p}_{k+1} - \mathbf{p}_k. \tag{10}$$

Given the video frame rate f (frames/s), pixel velocity is $\mathbf{v}_{px/s} = \Delta\mathbf{p}_{px}\, f$. Using a calibrated scale S (meters/pixel), this is converted to metric velocity:

$$\mathbf{v}_{m/s} = \mathbf{v}_{px/s}\, S = \Delta\mathbf{p}_{px}\, f\, S. \tag{11}$$

To reduce sensitivity to noisy tracks, robust statistics are computed along the dominant flow direction (estimated via PCA), reporting median speed and interquartile range (IQR) rather than relying only on a mean.

RANSAC for Robust Camera Motion (Planar Visual Odometry). Camera motion can corrupt the measured surface flow. To compensate, a planar visual odometry (VO) estimate is computed from *static* features outside the water ROI. A 2D motion model is fit between consecutive frames:

$$\mathbf{p}_{k+1} \approx A\mathbf{p}_k + \mathbf{t}, \tag{12}$$

where A is a 2×2 linear transform (allowing small rotation/scale) and $\mathbf{t} = [t_x\ t_y]^T$ is translation in pixels. Because feature tracks contain outliers (mismatches, moving objects, reflections), parameters are estimated using RANSAC: repeated random sampling is used to propose models, and the model with the largest inlier set under a reprojection threshold is selected. The resulting translation $\mathbf{t}$ is interpreted as the observed static-scene motion, and camera velocity in the rectified plane is approximated by its negative:

$$\mathbf{v}_{cam} \approx -\mathbf{t}\, f\, S. \tag{13}$$

Finally, the absolute water surface velocity is estimated by combining the relative water motion (from tracer flow) with the estimated camera motion:

$$\mathbf{v}^{abs}_{water} \approx \mathbf{v}^{rel}_{water} + \mathbf{v}_{cam}. \tag{14}$$

5.6 Experimental Setup and Results

Experiments were conducted on a short monocular video captured from a point-of-view (POV) perspective of a kayaker on a calm river. The camera exhibits noticeable translational and rotational motion, making direct velocity estimation from raw optical flow challenging.

Due to the lack of formal data for testing a youtube video of a kayaker was selected and then cropped to 10 seconds. This was done due to the motion of the gopro camera being very erratic. In the end a 10 second snippet was picked where the camera experinced only motion from the water and not from the human holding it.

After calibration, optical flow vectors were computed within the rectified region of interest using a Lucas–Kanade feature tracking approach. The resulting displacement vectors were converted into physical velocity estimates using the calibrated scale factor.

Figure 6 shows a representative output frame from the pipeline. Flow vectors are overlaid on the rectified image, and a summary statistic (median surface velocity) is displayed. For segments where feature tracking remained stable, the estimated velocities exhibited coherent directionality aligned with the visible river flow.

Figure 6: Representative output frame showing optical flow vectors overlaid on the rectified river surface.

6 Discussion and Limitations

While the proposed pipeline demonstrates the feasibility of estimating water surface velocity from monocular video, several practical limitations were observed.

In particular, the method was sensitive to image artifacts, motion blur, and inconsistent frame timing. Compression artifacts and rapid camera motion frequently degraded feature tracking performance, resulting in choppy or unreliable velocity estimates. Consequently, meaningful results were obtained only for short portions of the video where sufficient visual texture and temporal stability were present.

7 Conclusion and Future Work

This study investigated a computer-vision-based pipeline for estimating river surface velocity from monocular video. By combining planar homography rectification, scale calibration, and optical flow tracking, the system was able to recover plausible velocity estimates in selected segments of real-world footage.

Future work will focus on improving robustness through camera motion estimation using visual odometry, incorporating inertial sensors for stabilization, and evaluating the approach on higher-quality datasets with known ground truth measurements.

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